

Presupposition filtering in disjunction: The role of exclusive interpretation¹

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Abstract. Kalomoiros and Schwarz (2024) report symmetric presupposition projection across disjunction, a key datum for theories of projection. This study further asks whether projection across disjunctions is affected by exclusive interpretations. We observe that many fully semantic implementations of exclusivity and presupposition projection converge on the same, previously untested prediction: exclusive disjunction yields uniform projection (i.e., no filtering for either disjunct), so increasing exclusivity should reduce filtering. Mayr & Romoli (2016a) derive this prediction for Strong Kleene semantics when exclusivity is computed via bivalent exhaustification, and we show that it also follows, under the same assumption, for George’s (2008) algorithm, and for dynamic semantics (Rothschild, 2011). We also derive this prediction for strong-negation variants of trivalent exhaustification (Spector & Sudo, 2017). We then test this prediction in a Mandarin acceptability-judgment experiment adapted from Kalomoiros and Schwarz (2024), manipulating the rate of exclusive readings via environmental monotonicity. In fact, we find no evidence that increasing exclusivity affects filtering. If this null effect proves robust, it puts pressure on most of the above semantic implementations and appears compatible with only a weak-negation version of trivalent exhaustification that has been argued to be undesirable for independent reasons (Spector & Sudo, 2017), so we discuss possible future paths.

Keywords: disjunction, exclusive disjunction, presupposition projection, scalar implicature, exhaustification, trivalent semantics, dynamic semantics, experimental semantics.

1. Introduction

Increasing recent attention has been paid to the (a)symmetries of presupposition projection and filtering across binary connectives, given their theoretical importance for the architecture of dynamic semantic theories and related approaches to projection (Mandelkern et al, 2020; Kalomoiros & Schwarz, 2024; Loder et al., 2024, among many others). For presupposition projection across disjunction in particular, we highlight a potential confound that has not been systematically investigated: scalar implicatures and the resulting exclusive interpretations. Much recent work effectively assumes that implicatures are irrelevant for projection, so disjunction is treated as inclusive when computing presupposition projection (see Kalomoiros & Schwarz 2024, fn. 8), an assumption that is also implicit in many earlier accounts. We evaluate this assumption theoretically and experimentally by asking whether exclusive readings of disjunction affect presupposition projection.

We first explore theoretical predictions about the interaction between scalar implicatures and presupposition projection, focusing (for space, not principle) on accounts that implement both

¹ We would like to thank Mingming Liu for invaluable feedback on conceptualizing this project; Gennaro Chierchia, Jacopo Romoli, Alexandros Kalomoiros, Florian Schwarz, and Lisa Hofmann for helpful discussion; and members of the Harvard Meaning & Modality Lab, as well as audiences at Tsing Ch’a Sessions, the 11th Tsinghua Logic Colloquium, XPRAG 2025, and SuB30, for helpful feedback. All errors are our own.

computations in compositional semantics. We find that most such implementations predict that when a disjunction receives an exclusive implicature, presuppositions should uniformly project from both disjuncts, with no filtering. If this is right, then increasing the rate of exclusive readings should reduce the rate of presupposition filtering.

To test this prediction, we conducted an experiment in Mandarin, using an acceptability judgement paradigm modified from K&S (2024). We implemented this in Mandarin as a sample of convenience, given no reason to expect significant cross-linguistic variations in this regard. In fact, we find no evidence for an effect of exclusive readings on projection. If our manipulation of exclusive implicatures is correct, such results are in tension with most of the theoretical implementations we examined, and seem to be consistent with only a weak version of trivalent exhaustification that has been argued to be undesirable for independent reasons (Spector & Sudo, 2017). While a first step, if the empirical patterns in our experiment are found robust in future experimentation, they pose a challenge to theoretical approaches that account for both exclusive implicature and presupposition projection in compositional semantics.

The paper is organized as follows. Section 2 provides background on presupposition projection and our terminology. Section 3 outlines the theoretical landscape and derives predictions for disjunction under several semantic implementations of exclusive implicature and presupposition projection. Section 4 presents our experiment on Mandarin *huozhe* (‘or’) across different monotonic environments, reporting results on whether exclusivity affects filtering and, for comparison with prior work, on the (a)symmetry of filtering. Section 5 discusses theoretical implications. Section 6 concludes.

2. Backgrounds on presupposition projection

Presuppositions are implications of a sentence that are typically felt to be backgrounded and “taken for granted”, which is commonly diagnosed by embedding triggers under entailment-cancelling operators (e.g., negation, question, conditional antecedent) and observing that the presupposition tends to persist (Chierchia & McConnell-Ginet, 2000). They can be introduced by certain lexical items or constructions that are called *presupposition triggers* (e.g., “too,” “stop,” “know,” possessives). The “taken-for-granted” aspect is often implemented as a felicity/definedness requirement: the discourse context must already support the presupposition for the utterance to be fully acceptable.

Theoretical importance of presupposition projection across disjunction Presuppositions are theoretically significant for semantics and pragmatics because they are a classic pressure point for how (and where) to encode context-sensitivity in a theory of meaning, as in the classic debate involving definite descriptions (Russell, 1905; Strawson, 1950). Stalnaker’s (1973) influential framing makes the connection to pragmatics explicit by modeling presuppositions in terms of constraints on the common ground, conditions on when an assertion is appropriate given what interlocutors mutually accept. For theories of presuppositions in semantics, the central explanatory task is compositional: to predict the presupposition of a complex sentence from the presuppositions of its parts, i.e., to solve *the projection problem*.

A fundamental case for the projection problem is binary connectives. Order effects in conjunction and conditionals have played a major role in motivating dynamic semantics and

local-context approaches (Heim, 1983; Schlenker, 2009), which compute presupposition satisfaction relative to incrementally updated contexts rather than only the global common ground. Disjunction is particularly important because the update seems available in either direction without the order sensitivity found for conjunction. This contrast puts pressure on accounts that predict a uniform role for linear order across connectives.

Recent endeavors have been made to both clarify the empirical patterns and derive an explanatory theoretical approach to projection across connectives (Mandelkern et al, 2020; Kalomoiros & Schwarz, 2024; among many others). Our study aims to bring exclusive interpretation (often analyzed as scalar implicature) into this picture, since it is both a potential confound for projection judgments and bears on how theories of scalar implicatures (or exhaustification) interact with projection (e.g., Spector & Sudo, 2017; Marty & Romoli, 2021).

Terminology When the presupposition of a constituent clause is inherited by the larger clause, and thereby imposes a felicity condition on its context, that presupposition *projects*. Otherwise, it is *filtered*. (When a presupposition projects but is not licensed by its context, it may be *accommodated* either globally or locally, but accommodation is expected to incur a cost).

- (1) **Context:** We don't know whether John has a violin.
- a. # John's violin is expensive.
 - b. # John's violin is expensive, and he has a good violin.
 - c. ✓ John has a good violin, and his violin is expensive.

For example, (1a) presupposes that John has a violin. This presupposition projects in (1b), but does not project in (1c), given the felicity of (1c) in the context that is explicitly ignorant about this presupposition. Moreover, the presupposition of the right clause in (1c) is standardly taken to be filtered by information in the left clause, called *left-to-right filtering* (L-to-R filtering). *Right-to-left filtering* (R-to-L filtering) is the mirror pattern, where material in the right clause would license a presupposition in the left clause. (1b) is commonly taken to lack such R-to-L filtering, based on both intuitions and recent experimental results.

Filtering is *symmetric* if it is equally strong in both directions (as reflected in behavior measures), and *asymmetric* otherwise. Finally, we distinguish two symmetric endpoints: *uniform filtering*, where filtering is at ceiling in both directions, and *uniform projection*, where there is no evidence of filtering in either direction.

3. Theoretical analysis

It has long been debated whether scalar implicatures and presupposition projection are computed in compositional semantics or derived via pragmatic reasoning. Depending on where each component is located, we can distinguish four interface architectures:

- (a) **Scalar implicatures in semantics + presupposition projection in semantics**
- (b) Scalar implicatures in pragmatics + presupposition projection in semantics
- (c) Scalar implicatures in semantics + presupposition projection in pragmatics
- (d) Scalar implicatures in pragmatics + presupposition projection in pragmatics.

For purposes of this paper we restrict our attention to only (a), not as a claim about the correct architecture in general, but because we see they make some interesting predictions. Concretely, we analyzed predictions on disjunction arising from combining exhaustification-based semantic theories of exclusive implicatures with several semantic theories of presupposition projection. For exclusive implicature, we considered bivalent EXH (Fox, 2007; Chierchia et al., 2012), and two versions of trivalent EXH (Spector & Sudo, 2017): EXH¹ and EXH², respectively based on weak negation and strong negation. For presupposition projection, we considered strong Kleene semantics, George (2008)’s algorithm, and classic dynamic semantics.

3.1. Bivalent EXH as the source of exclusive implicature

For exclusive interpretations of disjunction, we first discuss the implications of a widespread approach regarding EXH (Fox, 2007; Chierchia et al., 2012; among many others). When applied to a prejacent, EXH asserts it and negates all its innocently excludable (*IE*) alternatives, as defined in (2), which we adopted from Mayr and Romoli (2016a, p.13):

- (2) a. $Exh_{Alt(p)}(p) = p \wedge \forall q \in IE(p, Alt(p))[\neg q]$
 b. $IE(p, Alt(p)) = \{q \in Alt(p) : p \not\sqsubseteq q \wedge \neg \exists r[r \in Alt(p) \wedge (p \wedge \neg q) \subseteq r]\}$

For a disjunction $\phi \vee \psi$, the standard alternatives are given in (3a). Under the definition in (2b), only the conjunction alternative is innocently excludable, shown in (3b). As a result, EXH strengthens $\phi \vee \psi$ to an exclusive interpretation in (3c):

- (3) a. $Alt(\phi \vee \psi) = \{\phi, \psi, \phi \wedge \psi\}$
 b. $IE(\phi \vee \psi, Alt(\phi \vee \psi)) = \{\phi \wedge \psi\}$
 c. $Exh_{Alt(\phi \vee \psi)}(\phi \vee \psi) = (\phi \vee \psi) \wedge \neg(\phi \wedge \psi)$

In the rest of §3.1, we examine how the exclusive reading produced by bivalent EXH affects presupposition projection in disjunction. We assume Exh scopes over disjunction, as in (3c). In general, we treat the exclusive implicature generated by EXH as relevant for computing presupposition projection, which requires a nonstandard architectural choice given the scope configuration in (3c). However, the following three assumptions seem incompatible: (a) EXH is bivalent, so it cannot handle undefinedness; (b) the LF (with EXH scoping over the disjunction) is interpreted hierarchically in the strict sense; (c) at least one disjunct contains a presupposition trigger. With (c), both the disjunction and its conjunction alternative could be undefined, so following the strict hierarchical assumption in (b), a bivalent EXH in (a) would have no way to handle this situation. To keep the derivation well-defined, we relax (b) and allow the exclusive strengthening to be computed in a way that can feed the presupposition projection computation for disjunction.

3.1.1. Bivalent EXH + Strong Kleene semantics

Figures 1 and 2 give Strong Kleene truth tables for inclusive vs. exclusive disjunction. Based on these truth tables, Mayr and Romoli (2016a) pointed out that Strong Kleene semantics predicts a conditional presupposition for a disjunction interpreted *inclusively*, but a non-

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conditional presupposition for a disjunction interpreted *exclusively*. In other words, there will be uniform filtering for an inclusive disjunction, but no filtering for an exclusive disjunction.²

$\vee$	1	0	#
1	1	1	1
0	1	0	#
#	1	#	#

Figure 1 Strong Kleene truth table: Inclusive disjunction

$\underline{\vee}$	1	0	#
1	0	1	#
0	1	0	#
#	#	#	#

Figure 2 Strong Kleene truth table: Exclusive disjunction

3.1.2. Bivalent EXH + George's (2008) algorithm

George (2008)'s algorithm is a principled procedure to determine the trivalent truth table based on the classic truth table of a connective. It succeeds in generating only left-to-right filtering for conjunction but uniform filtering for (inclusive) disjunction. We observe that it predicts the same trivalent truth table as Strong Kleene semantics for an exclusive disjunction.

Algorithm 1 (adopted from K&S 2024, p.365) shows a simplified version of this procedure.

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Given ( $\alpha * \beta$ ) (where  $*$  is a binary connective), consider first  $\alpha$ 
  if on the basis of the truth value of  $\alpha$  and the classical semantics of the  $*$  connective,
  you can assign a truth value to the whole sentence, then
    | do so;
  else
    if there exists a possible truth value for  $\beta$  that can make the sentence True on the
    classical truth table, then
      check the value of  $\beta$ :
        if the classical truth table assigns a value to sentence on the basis of the value
        of  $\alpha$  and  $\beta$  then
          | Assign that value to the whole sentence;
        else
          | assign # to the whole sentence;
        end
      end
    else
      | assign # to the whole sentence;
    end
  end
end

```

Algorithm 1: The algorithm of George (2008)

For an exclusive disjunction, when α is #, the classical truth conditions of exclusive “or” do not allow a determinate truth value for the whole sentence, and there is no possible truth value for β that can make the sentence *True* given the uncertainty about α . The algorithm therefore assigns # to the whole sentence. Similarly, when α is *True* or *False* but β is #, there is still insufficient information to assign a classical value, and the algorithm again returns #. Therefore, George (2008) also predicts no filtering for an exclusively interpreted disjunction.

² It is clear that Weak and Middle Kleene semantics would make the same prediction for exclusive disjunction, because they would generate the same truth table.

3.1.3. Bivalent EXH + dynamic semantics

We further observe that classic dynamic semantics predicts the same – the local contexts for both disjuncts of an exclusive disjunction is the global context. Assuming the restrictions on the metalanguage of dynamic semantics provided in Rothschild (2011), there are only eight possible ways to define the CCP of an exclusive disjunction.³ Figure 3 shows all eight possible CCPs for an exclusive disjunction $\alpha \text{ xor } \beta$.

(a) $C[\alpha][\neg\beta] \cup C[\beta][\neg\alpha]$	(e) $C - C[\alpha][\beta] - C[\neg\beta][\neg\alpha]$
(b) $C[\neg\alpha][\beta] \cup C[\neg\beta][\alpha]$	(f) $C - C[\neg\alpha][\neg\beta] - C[\beta][\alpha]$
(c) $C[\alpha][\neg\beta] \cup C[\neg\alpha][\beta]$	(g) $C - C[\alpha][\beta] - C[\neg\alpha][\neg\beta]$
(d) $C[\beta][\neg\alpha] \cup C[\neg\beta][\alpha]$	(h) $C - C[\beta][\alpha] - C[\neg\beta][\neg\alpha]$

Figure 3 All possible context change potentials for $C[\alpha \text{ xor } \beta]$

Turning to their definedness conditions:

For (a), $C[\alpha]$ and $C[\beta]$ need to be defined, meaning there is no filtering for α and β .

For (e), $C[\alpha]$ and $C[\neg\beta]$ need to be defined. Since $C[\neg\beta] = C - C[\beta]$, this means $C[\beta]$ should also be defined, equating the situation with (a). (b) and (f) are also similar to (a).

For (c), $C[\alpha]$, $C[\alpha][\neg\beta]$, and $C[\neg\alpha][\beta]$ should be defined. Since $C[\alpha][\neg\beta] = C[\alpha] - C[\alpha][\beta]$, it requires the definedness of $C[\alpha][\beta]$. Thus, β should be defined on both $C[\alpha]$ and $C[\neg\alpha]$, i.e., the entire context set C . Thus, there is no filtering for both α and β in (c). The cases for (g), (d), and (h) are similar to (c).

Therefore, in an exclusive disjunction, dynamic semantics predicts that the local contexts for both disjuncts are the global context, so there is no filtering.

Interim summary of §3.1 If exclusive readings are derived via bivalent exhaustification as in (3c), then across the three semantic implementations of projection considered here (Strong Kleene, George’s algorithm, and dynamic semantics), an exclusively interpreted disjunction is predicted to allow uniform projection (i.e., no filtering). Therefore, these implementations converge on the same prediction: increasing the rate of exclusive readings of disjunction should reduce presupposition filtering.

3.2. Trivalent EXH as the source of exclusive implicature

Recently, many papers have explored different trivalent versions of EXH (Spector & Sudo, 2017; Marty & Romoli, 2021; among many others), partly to address phenomena often discussed under “presupposed ignorance” / “presupposed scalar implicatures.” This paper will discuss the implications of trivalent EXH for presupposition projection itself, in terms of disjunction. We restrict our attention to EXH¹ and EXH² proposed in Spector and Sudo (2017), and remark that the trivalent EXH in Marty and Romoli (2021) inherits the predictions of EXH² in terms of presupposition projection.

³ Thanks to an anonymous reviewer for pointing out the required assumptions on the metalanguage of CCP here.

Throughout §3.2, we assume that the presuppositions of the prejacent and its alternatives are computed in the usual way (by whatever semantic theory of projection one adopts) before applying EXH¹/ EXH². The specific choice of semantic projection theory will not matter here.

3.2.1. EXH¹ + any semantic theory for presupposition projection

EXH¹ is based on the weak negation of IE-alternatives, yielding *True* when all IE-alternatives are either *False* or *undefined*. Since it can return *True* even when the IE-alternatives are undefined, it does not inherit presuppositions from those alternatives. (4) and (5) reproduce the definition (29) and its paraphrase (30) from Spector & Sudo (2017, p. 485).

(4) Definition of EXH¹

$$\begin{aligned} & \llbracket \text{EXH}_{\text{Alt}(\phi)}^1 \phi \rrbracket(w) \\ &= \begin{cases} \# & \text{iff } \llbracket \phi \rrbracket(w) = \# \\ 1 & \text{iff } \llbracket \phi \rrbracket(w) = 1 \text{ and for all } \psi \in \text{IE}^1(\phi, \text{Alt}(\phi)), \llbracket \psi \rrbracket(w) = 0 \text{ or } \# \\ 0 & \text{iff } \llbracket \phi \rrbracket(w) = 0 \text{ or for some } \psi \in \text{IE}^1(\phi, \text{Alt}(\phi)), \llbracket \psi \rrbracket(w) = 1 \end{cases} \end{aligned}$$

(5) Equivalent reformulation of EXH¹

- $\text{EXH}_{\text{Alt}(\phi)}^1 \phi$
- a. asserts ϕ and $\sim\psi$ of all alternatives $\psi \in \text{IE}^1(\phi, \text{Alt}(\phi))$; and
 - b. presupposes whatever ϕ presupposes.

The key contrast between bivalent EXH and EXH¹ for projection concerns their difference in presupposition tolerance. For bivalent EXH, we have to assume strengthening of disjunction via EXH precedes computations of presupposition and its projection (discussed in §3.1). By contrast, trivalent EXH¹ (and EXH²) is presupposition-tolerant, so computation of presupposition projection across disjunction can precede exhaustification, following the scope relation $\text{Exh}_{\text{Alt}(\phi \vee \psi)}^1(\phi \vee \psi)$. Under this ordering, trivalent EXH can affect projection only by importing presuppositions from the IE-alternatives. Since EXH¹ does not inherit presuppositions from alternatives, it predicts no modulation of presupposition projection.

3.2.2. EXH² + any semantic theory for presupposition projection

EXH² is based on strong negation of IE-alternatives: it returns *True* only when the IE-alternatives are all *False*, and returns $\#$ when some IE-alternative is undefined. Consequently, EXH² *does* inherit (or “re-introduce”) presuppositions from IE-alternatives. (6) and (7) reproduce the definition (63) and its paraphrase (64) from Spector & Sudo (2017, p. 498).

(6) Definition of EXH²

$$\begin{aligned} & \llbracket \text{EXH}_{\text{Alt}(\phi)}^2 \phi \rrbracket(w) \\ &= \begin{cases} \# & \text{iff } \llbracket \phi \rrbracket(w) = \# \text{ or for some } \psi \in \text{IE}^2(\phi, \text{Alt}(\phi)), \llbracket \psi \rrbracket(w) = \# \\ 1 & \text{iff } \llbracket \phi \rrbracket(w) = 1 \text{ and for all } \psi \in \text{IE}^2(\phi, \text{Alt}(\phi)), \llbracket \psi \rrbracket(w) = 0 \\ 0 & \text{iff } \llbracket \phi \rrbracket(w) = 0 \text{ or for some } \psi \in \text{IE}^2(\phi, \text{Alt}(\phi)), \llbracket \psi \rrbracket(w) = 1 \\ & \text{and for no } \psi \in \text{IE}^2(\phi, \text{Alt}(\phi)), \llbracket \psi \rrbracket(w) = \# \end{cases} \end{aligned}$$

(7) Equivalent reformulation of EXH²

$\text{EXH}_{\text{Alt}(\phi)}^2 \phi$

- a. asserts ϕ and the strong negation of all alternatives $\psi \in \text{IE}^2(\phi, \text{Alt}(\phi))$;
- b. presupposes whatever ϕ presupposes and the negated alternatives ψ presuppose.

For disjunction, the relevant IE-alternative is the conjunction: $\text{IE}(\alpha \vee \beta) = \{\alpha \wedge \beta\}$ (cf. (3b)). Given (7b), EXH² therefore yields a presupposition for the whole parse that includes the presupposition of $\alpha \wedge \beta$, in addition to that of $\alpha \vee \beta$, as shown in (8), where we use $ps(\phi)$ to mean the presupposition of ϕ :

$$(8) \quad ps(\text{Exh}^2(\alpha \vee \beta)) = ps(\alpha \vee \beta) \wedge ps(\neg(\alpha \wedge \beta)) = ps(\alpha \vee \beta) \wedge ps(\alpha \wedge \beta)$$

Now suppose (for illustration) that β carries a presupposition while α does not. Under standard projection assumptions, $ps(\alpha \vee \beta)$ entails *at least* a conditional requirement ($\neg\alpha \rightarrow ps(\beta)$), while $ps(\alpha \wedge \beta)$ entails *at least* ($\alpha \rightarrow ps(\beta)$). Conjoining these yields $ps(\beta)$ outright. The mirror case where only α carries a presupposition is parallel. Thus, under EXH² the presupposition of either disjunct ends up projecting to the whole exhaustified disjunction. Therefore, if EXH² is the source of exclusive implicatures, increasing the rate of exclusive implicatures of disjunctions would reduce filtering, matching the prediction from §3.1.

3.3. Summary of theoretical analysis

Taken together, the semantic implementations of exclusive interpretation and presupposition projection we considered fall into two prediction classes in (9a) and (9b):

(9) **Theories categorized by their predictions**

a. **Type A theories**

Prediction: increasing the rate of exclusive interpretation of disjunctions reduces presupposition filtering

- i. Bivalent EXH + Strong Kleene semantics
- ii. Bivalent EXH + George (2008)’s algorithm
- iii. Bivalent EXH + classic dynamic semantics
- iv. EXH² + any semantic theory for presupposition projection

b. **Type B theories**

Prediction: increasing the rate of exclusive interpretation of disjunctions has no effect on presupposition filtering

- i. EXH¹ + any semantic theory for presupposition projection

4. Experiment: exclusive interpretation on presupposition projection

Background K&S (2024) reported symmetric projection judgments for English *either...or* and interpreted this symmetry as uniform filtering, but note a reviewer’s concern (K&S, 2024: fn. 8) that this symmetry could be influenced by the availability of exclusive interpretations in the sense of Mayr & Romoli (2016a), potentially making an “uniform filtering” pattern hard to distinguish from “uniform projection.” Although K&S argued against the uniform-projection

interpretation (fn. 24), the potential role of exclusivity in presupposition projection across disjunction has not been directly tested. Our study targets this question by testing whether increased exclusivity reduces filtering, as predicted by the Type A vs. Type B implementations surveyed in §3.

4.1. Design: “huozhe” in UE and DE environments

Our study tested this question in Mandarin, for sample convenience given no particular expectations of cross-linguistic variation (although that is always a possibility). We used the single-particle disjunction *huozhe* (‘or’) and manipulated the availability of exclusive readings.

Within-subject design We used a 2×2 within-subject design adopted from K&S (2024) to diagnose projection/filtering, crossing two factors, PREDICATE (Presuppositional (PS) vs. Non-presuppositional (NOPs)) with ORDER (FIRST vs. SECOND), yielding four conditions: PSFIRST, PSSECOND, NOPSFIRST, and NOPSSSECOND. The label “PS” means the test sentence contains a presupposition trigger; while the label “NOPs” means the test sentence replaces that presuppositional trigger with a suitable non-presuppositional predicate. The label “FIRST” means the presupposition trigger (or its non-presuppositional counterpart) is in the first disjunct; the label “SECOND” means the trigger (or its NOPs counterpart) is in the second disjunct. To prevent global accommodation, test sentences in all these four critical disjunction conditions are preceded by a CONTEXT that is “explicitly ignorant” (EI) about the relevant presupposition. In (10)-(12) below, we use examples translated from our experimental stimuli to illustrate the context and test sentences in different conditions.

(10) CONTEXT (that precedes a test sentence)

a. **Explicit Ignorant (EI):**

I didn’t know Li at all before, and I didn’t know if he ever drank. At tonight’s dinner gathering, most people drank, but Li didn’t drink a drop, so I thought:

b. **Supportive (S):**

Li is my college classmate. I know that back in college he liked drinking, but after we graduated we lost touch. At tonight’s dinner gathering, most people drank, but Li didn’t drink a drop, so I thought:

(11) **Test sentences** (in critical disjunction conditions)

- a. PSFIRST: Li has quit drinking or never drank.
- b. PSSECOND: Li never drank or has quit drinking.
- c. NOPSFIRST: Li dislikes drinking or never drank.
- d. NOPSSSECOND: Li never drank or dislikes drinking.

We also included two additional control conditions following K&S: EISIMPLEPS, and SSIMPLEPS. The labels “EI” and “S” in the beginning respectively represent whether the preceding CONTEXT is *explicitly ignorant*, or *supportive* of the relevant presupposition (as shown in (10) above). The label “SIMPLE” means that in the test sentences, presupposition triggers are *directly* embedded in the antecedent of conditionals (as shown in (12) below), which are assumed to be presupposition holes. Thus, the difference of ratings between EISIMPLEPS and SSIMPLEPS establishes the deduction in rating caused by presupposition projection (when there is only local accommodation as rescue).

(12) **Test sentence** (in control conditions)

SIMPLEPS: If Li has quit drinking, then his behavior tonight makes total sense.

Between-subject design To manipulate exclusivity, we added a third factor as a between-subject manipulation: Environmental MONOTONICITY, with two levels (downward entailing (DE) vs. upward entailing (UE)). Each participant was assigned to exactly one environment (DE or UE), and all critical disjunction trials they judged were presented in that environment. We expect disjunctions in a UE environment to have a higher rate of exclusive implicatures than those in a DE environment.

To verify that the MONOTONICITY manipulation modulates exclusivity for *huozhe* ('or'), we conducted a separate norming task ($n = 30$) in which participants judged both UE and DE versions of the norming materials (within-subject). Responses were binary and were coded as *exclusive* when participants judged a disjunctive prediction (in a UE environment) to be incorrect in a "both true" scenario (see (13)). In this norming task, exclusive responses were more frequent in UE than in DE (UE: 23.3%; DE: 0%), confirming that the monotonicity manipulation increases exclusivity (Fisher's exact test, $p = .011$).

(13) **Norming task** (a trial in a UE environment):

(a) **Participant view:**

投票单选

* 04 小李说: “我相信小张会来或者小王会来。”

事实上, 后来小张和小王都来了。

请问你觉得小李的预测是否正确?

☐ 正确

☐ 错误

(b) **Translation:**

Li said: “I believe Zhang will come or Wang will come.”

In fact, afterwards both Zhang and Wang came.

Do you consider Li's prediction correct or incorrect?

(c) **Glossing** (of the quoted sentence):

Wo	xiangxin	xiaozhang	hui	lai	huozhe	xiaowang	hui	lai
I	believe	Zhang	will	come	or	Wang	will	come

In the main experiment, the UE environment was implemented as an unembedded disjunction, whereas the DE environment was implemented by embedding the same disjunction in the antecedent of a conditional (see (14) for an illustration of translated stimuli).

(14) **Test sentences** (in PSFIRST condition) in UE and DE environments:

- UE:** Li has quit drinking or never drank.
- DE:** If Li has quit drinking or never drank, then his behavior tonight makes total sense.

Following prior work, the disjunctions in our critical items are *bathroom disjunctions*, where the negation of one disjunct is equivalent to the presupposition of the other disjunct (so-called for Barbara Partee’s example: *Either it’s in a weird place or this house has no bathroom*). This structure ensures that filtering is in principle available under an inclusive reading of the disjunction, which is crucial for testing our projection predictions.⁴

4.2. Stimuli

We selected two effective presupposition triggers, *jie* (‘quit’) and *zhidao* (‘know’), based on pilot data showing ratings for these in the EISIMPLEPS condition were significantly lower than in SSIMPLEPS. The non-presuppositional counterpart predicates in NOPS conditions for *jie* and *zhidao* are respectively *buxihuan* (‘dislike’) and *juede* (‘think’).

We created 2 items for each trigger. Each item included six within-subject conditions: four critical disjunction conditions (PSFIRST/PSSECOND/NOPSFIRST/NOPSSECOND) and two control conditions (EISIMPLEPS/SSIMPLEPS). We used a pseudo-Latin square design to ensure that each participant saw exactly one critical disjunction condition and one control condition per item; this yielded four counterbalanced lists within each MONOTONICITY group (UE/DE), eight in total. Each list has 15 trials: 8 target trials (2 per item), 2 catch trials, 4 filler trials, and 1 inferential task. A target trial consisted of: (i) an either Explicitly Ignorant (EI) or Supportive (S) context; (ii) a test sentence involving a trigger or its NOPS counterpart; (iii) a judgment task for the naturalness of the test sentence in the context on a 7-point scale, as in (15).

(15) Example trial (PSFIRST, DE; trigger *jie* ‘quit’)

(a) Participant view:

*8. 我之前完全不认识小李，不知道他有没有喝过酒。在今晚的饭局上，大多数人都喝酒了，但小李滴酒不沾，于是我想：
如果小李已经戒酒了或者从来不喝酒，那么他今晚的行为很合理。

请给划线句子在语境中的自然程度打分：

完全不自然

完全自然

①

②

③

④

⑤

⑥

⑦

(b) Translation:

I didn’t know Li at all before, and I didn’t know if he ever drank. At tonight’s dinner gathering, most people drank, but Li didn’t drink a drop, so I thought:
If Li has quit drinking or never drank, then his behavior tonight makes total sense.

Please rate how natural the underlined sentence is in the context:

[Endpoints labeled: 1 “completely unnatural” – 7 “completely natural”]

⁴ Thanks to an anonymous reviewer for pointing out that the use of bathroom disjunctions might raise concerns about the status of the conjunction alternative, and thus about whether an exclusive implicature is generated. We return to this potential limitation in §4.6; for present purposes, we assume that the monotonicity-based UE > DE contrast in exclusivity established by the norming task carries over to the bathroom-disjunction materials used in the main experiment.

(c) **Test sentence glossing** (of the underlined sentence in (15a)):

ruguo xiaoli yijing jie jiu le huozhe conglai bu he jiu,
if Li already quit alcohol PFV or always NEG drink alcohol,
name ta jinwande xingwei hen heli.
then he tonight-DE behavior very reasonable.
'If Li has quit drinking or never drank, then his behavior tonight makes total sense.'

The full set of stimuli, together with the R scripts of statistical analysis for §4.5, will be accessible at the OSF page for this paper, at <https://osf.io/5n98y/>.

4.3. Predictions

We first expect the paradigm to be sensitive to presuppositional definedness requirements in Mandarin when filtering is unavailable. In particular, in the **control** conditions (EISIMPLEPS vs SSIMPLEPS), the CONTEXT manipulation should have a main effect.

Turning to **critical** disjunction conditions (PSFIRST/PSSECOND/NOPSFIRST/NOPSSECOND), if exclusivity affects filtering, this should surface as a PREDICATE × MONOTONICITY interaction: the PS–NOPS contrast should differ between UE and DE environments.

Finally, in terms of filtering (a)symmetries, symmetric filtering predicts no PREDICATE×ORDER interaction. If the symmetry reflects *uniform filtering* rather than *uniform projection*, PS conditions should be as acceptable as NOPS conditions, yielding no main effect of PREDICATE.

4.4. Participants and Procedure

The experiment was administered online via Wenjuanxing. We recruited 141 self-identified native Mandarin speakers via university student WeChat groups. Each participant was randomly assigned to one of the eight counterbalanced lists (four per MONOTONICITY environment: UE vs DE) generated by the pseudo-Latin square design described in §4.2, and trials were presented in randomized order. Participants received 3.6 RMB upon completion. We excluded 21 participants who failed at least one catch trial, leaving 120 for analysis.

4.5. Results

We analyzed naturalness ratings with cumulative-link mixed-effects models with a logit link.⁵ Unless noted otherwise, the primary models exclude data from the two control conditions and focus on the critical disjunction conditions (PSFIRST/PSSECOND/NOPSFIRST/NOPSSECOND).

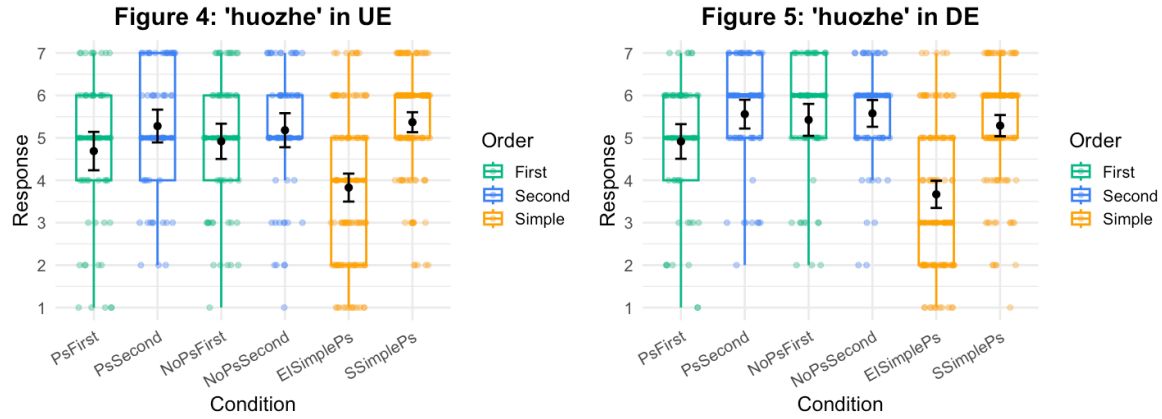
Effectiveness of paradigm Restricting the data to only the two control conditions (EISIMPLEPS, SSIMPLEPS), CONTEXT reliably affected ratings: sentences were degraded under

⁵ We used treatment coding. Reference levels are: PREDICATE = NoPs, ORDER = Second, MONOTONICITY = DE, TRIGGER = zhidao, CONTEXT = S.

Presupposition filtering in disjunction: The role of exclusive interpretation

explicit ignorance (EI) relative to supportive (S) contexts ($p < .001$), confirming that the paradigm detects presuppositional definedness costs in Mandarin when filtering is unavailable.

Exclusivity and filtering Figures 4 and 5 plot ratings by condition respectively for different MONOTONICITY (UE/DE).



Black dots show cell means; error bars show mean $\pm 1.96 \times SE$ (normal-approx. 95% CIs). Colored boxplots summarize median and IQR, with whiskers = $1.5 \times IQR$; jittered points are individual trial-level responses.

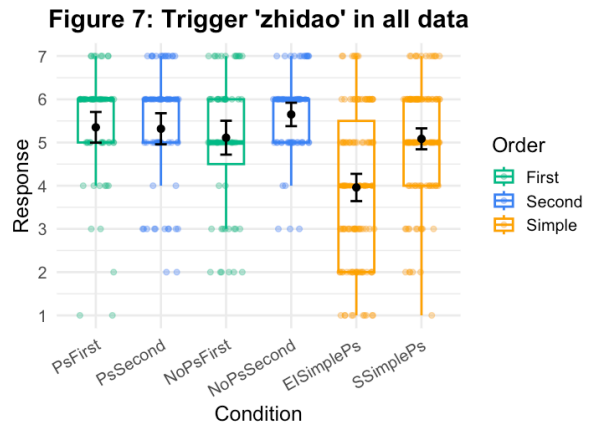
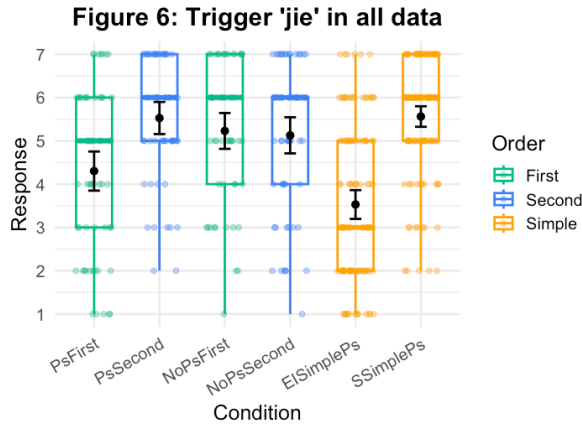
The central test for whether increased exclusivity modulates filtering was the PREDICATE $\times$ MONOTONICITY interaction. In the full CLMM model shown below,

$$\text{RATING} \sim \text{PREDICATE} \times \text{MONOTONICITY} \times \text{TRIGGER} + \text{PREDICATE} \times \text{ORDER} \times \text{TRIGGER} + (1 + \text{PREDICATE} + \text{ORDER} \mid \text{PARTICIPANT}) + (1 + \text{PREDICATE} + \text{ORDER} \mid \text{ITEM})$$

PREDICATE $\times$ MONOTONICITY was not a significant interaction ($p = .30$), and neither was the three-way interaction involving TRIGGER ($p = .81$). In our planned Bayesian model comparison, we fit the full model with homologous *brms* models with weakly informative priors, and compared it to a reduced model that omits the PREDICATE $\times$ MONOTONICITY term (while retaining MONOTONICITY $\times$ TRIGGER). We computed the Bayes factor in favor of the full model, and found a value of $BF_{10} = 0.52$, which corresponds to $BF_{01} \approx 1.92$. Thus, the Bayesian comparison provides weak evidence favoring the simpler model and no positive evidence for the PREDICATE $\times$ MONOTONICITY interaction.

Symmetry of filtering and trigger dependence To test whether filtering across disjunction is symmetric, we asked whether the presuppositional penalty depends on ORDER, and whether this dependency differs by presupposition TRIGGER. In the full model, the PREDICATE $\times$ ORDER $\times$ TRIGGER interaction showed a trend ($p = .053$), motivating trigger-wise follow-ups.

Figure 6 and 7 below show the ratings per condition respectively for each trigger (where plot elements and labels are the same as those in Figures 4–5), suggesting *asymmetry* for “jie” but *symmetry* for “zhidao,” which is also confirmed in following statistical tests.



Follow-up tests of the $\text{PREDICATE} \times \text{ORDER}$ interaction within each trigger (Bonferroni-corrected across triggers) showed a dissociation: for “jie” (quit), $\text{PREDICATE} \times \text{ORDER}$ was significant ($\beta = -1.81$, $\text{SE} = 0.72$, $p = .01$), indicating weaker right-to-left filtering than left-to-right. For “zhidao” (know), $\text{PREDICATE} \times \text{ORDER}$ was not significant ($p = .40$). To assess whether the apparent symmetry of “zhidao” reflects uniform filtering or uniform projection, we tested the main effect of PREDICATE within “zhidao” and found it insignificant ($p = .99$), which supports uniform filtering.

Baseline projection strength To test if there is also a trigger split in baseline projection strength, we fit a control-only CLMM (including only EISIMPLEPS and SSIMPLEPS conditions):

$$\text{RATING} \sim \text{CONTEXT} \times \text{TRIGGER} + (1 \mid \text{PARTICIPANT}) + (1 \mid \text{ITEM})$$

This model yields a robust CONTEXT effect ($\beta = -1.22$, $\text{SE} = 0.26$, $p < .001$) and a $\text{CONTEXT} \times \text{TRIGGER}$ interaction ($\beta = -1.62$, $\text{SE} = 0.41$, $p < .001$), indicating that the EI penalty differs by trigger. Follow-up tests of CONTEXT within each trigger (also Bonferroni-corrected across triggers) showed stronger EI penalty for “jie” ($\beta = -2.84$, $\text{SE} = 0.30$, $p < .001$) than “zhidao” ($\beta = -1.22$, $\text{SE} = 0.27$, $p < .001$), indicating stronger projection for “jie.”

Robustness checks Finally, our key qualitative conclusions, no $\text{PREDICATE} \times \text{MONOTONICITY}$ effect, and a $\text{PREDICATE} \times \text{ORDER}$ effect only for “jie,” were found to be stable across alternative random-effects specifications (simpler intercept-only vs. more complex structures).

4.6. Experimental summary and discussion

Exclusivity and filtering We found no evidence that environmental monotonicity (as a proxy for exclusivity) modulates presupposition filtering ($\text{PREDICATE} \times \text{MONOTONICITY}$ not significant; Bayes factor weakly favoring the simpler model without this interaction). Assuming our monotonicity manipulation successfully increases the rate of exclusive readings (see norming in §4.1), these results challenge Type A accounts (e.g., bivalent EXH + Strong Kleene semantics / George 2008 / classic dynamic semantics; and trivalent EXH² based on strong negation that reintroduces presuppositions), but are consistent with Type B accounts (e.g., trivalent EXH¹ based on weak negation that does not alter presuppositions).

(A)symmetry of filtering and baseline projection strength We found that, unexpectedly, (a)symmetry of filtering depends on the trigger. There is a significant $\text{PREDICATE} \times \text{ORDER}$

interaction for *jie* ('quit'), but not for *zhidao* ('know'), suggesting filtering is asymmetric for *jie* but symmetric for *zhidao*. Further, the lack of main effect of PREDICATE shows the symmetric filtering for *zhidao* is actually uniform filtering. Thus, while both L-to-R and R-to-L filtering are at ceiling for *zhidao*, R-to-L filtering is (available but) not at ceiling for *jie*.

We tentatively explain the trigger split in filtering (a)symmetry in terms of baseline projection strength, which sets something like a *maximal presuppositional penalty* available to be mitigated by filtering (operationally, the size of the NOPS–PS rating difference when filtering is unavailable). Such cross-trigger variation in projection strength and contextual requirement stringency is widely documented (e.g., Tonhauser et al., 2018; Wilcox et al., 2021; Wilcox, 2022; among many others). In our materials, the control conditions provide converging evidence for such a difference: when the triggers are directly embedded in a conditional antecedent, the CONTEXT manipulation (EI vs. S) yields a larger penalty for *jie* ('quit') than for *zhidao* ('know'), suggesting that *jie*'s presupposition projects more strongly and therefore sets a larger upper bound on the NOPS–PS gap. This qualitative profile mirrors Wilcox's (2022) finding that English change-of-state predicates show stronger projection effects than cognitive factives when embedded in conditional antecedents (p. 119).

Given this background, the disjunction results are compatible with a simple sensitivity-based hypothesis. In the critical disjunction conditions, assuming right-to-left filtering is available but not at ceiling, i.e., it mitigates most, but not all, of this gap, a small *residual* presuppositional penalty remains when the trigger is in the first disjunct and must be licensed by information in the second. Crucially, that residual will be detectable only if the baseline NOPS–PS gap is sufficiently large: for *jie* ('quit'), a strong baseline presuppositional penalty means that even a small remainder after partial right-to-left filtering can register as a PREDICATE $\times$ ORDER interaction, whereas for *zhidao* ('know'), a weaker baseline penalty implies a smaller remainder that may be too small to detect with two items per trigger and $N=120$, yielding an apparent pattern of uniform filtering. On this view, our data is consistent with a shared filtering profile across triggers (left-to-right filtering at ceiling; right-to-left filtering available but slightly weaker), but baseline projection strength determines whether the right-to-left shortfall becomes measurable. This possibility fits naturally with existing evidence for cross-trigger heterogeneity and motivates closer attention to sensitivity/ceiling effects in future tests of filtering (a)symmetry.

Limitations and methodological implications Two limitations are worth noting, with corresponding directions for future work. First, our critical stimuli are *bathroom disjunctions*, where the negation of one disjunct coincides with the other's presupposition. This structure makes the conjunction alternative systematically problematic (always false or undefined), which may weaken the incentive to negate it and attenuate exclusive implicature. Nevertheless, an independent norming task confirmed the intended $UE > DE$ contrast in exclusive responses, and it is reasonable to expect that at least some of this contrast carries over to the bathroom-disjunction materials.

Future studies can preserve the filtering logic while keeping the conjunction alternative well-formed by weakening the "bathroom" relation from equivalence to asymmetric entailment: let p be the presupposition-triggering disjunct with presupposition π , and choose the other disjunct q so that $\neg q$ entails π while q remains compatible with π (e.g., $p = \text{John quit smoking}$, $\pi = \text{John}$

used to smoke, $q = \text{John never smoked Marlboro}$). Then $p \wedge q$ is consistent, providing a cleaner target for exclusive strengthening.

Second, the trigger split between *jie* (‘quit’) and *zhidao* (‘know’) underscores a sensitivity issue for symmetry diagnostics. If baseline projection is weak and filtering is near ceiling, any residual penalty under right-to-left filtering may be too small to detect, producing “apparent symmetry” even when right-to-left filtering is slightly weaker than left-to-right filtering. Future work aimed at diagnosing filtering (a)symmetry should therefore increase item counts per trigger and sample triggers spanning a wider range of baseline projection strength, so that small order effects can be separated from ceiling/scale-compression limits.

5. Implications for alternative-based strengthening and presupposition projection

The empirical result and where it lands If we take our monotonicity manipulation to successfully modulate the likelihood of an exclusive reading (as supported by the norming results reported in §4.1), the main empirical takeaway of our experiment is that exclusive interpretation of disjunctions did not measurably affect presupposition filtering.

Recall in §3 we surveyed fully compositional implementations that combine (i) exclusive interpretations of disjunction (via exhaustification) and (ii) semantic theories of presupposition projection, and divided them based on their predictions into **Type A** systems, which predict increased exclusivity should *reduce* filtering (bivalent EXH with Strong Kleene, George-style procedures, or classic dynamic semantics; and strong-negation based trivalent EXH² that reintroduces presuppositions from alternatives), and **Type B** systems, which predict modulating exclusivity has *no effect* on filtering (weak-negation based EXH¹ which doesn’t import presuppositions from alternatives). The observed lack of interaction between exclusivity and filtering in disjunctions is compatible with Type B systems, while posing a challenge to the Type A class of implementations.

Why Type A systems are not straightforward to reconcile It is natural to ask whether Type A systems can be minimally adjusted to predict no effect of exclusivity on filtering. For Type A theories associated with bivalent EXH, the prediction that exclusivity eliminates filtering depends on a substantive *feed-forward* assumption: the strengthened (exclusive) truth conditions must be computed early enough to be visible to the projection computation for disjunction, so that definedness conditions/local context are computed for exclusive disjunction rather than inclusive disjunction. Denying this feed-forward access would be an obvious way to remove the interaction, but in §3.1 we adopted the feed-forward ordering to avoid a deeper mismatch between bivalent exhaustification and semantic approaches for projection in the first place; abandoning it reintroduces that problem rather than offering a simple repair. By contrast, the Type A theories associated with trivalent EXH² do not rely on exclusivity feeding projection; it relies on the reverse dependency. Projection/filtering is computed first (on an inclusive interpretation), and then EXH² yields uniform projection by importing presuppositions from the conjunction alternative, i.e., a feedback route from alternatives to presuppositional requirements. Since this mechanism is built into the strong-negation semantics of EXH² (as in (6)), reconciling EXH²-based Type A systems with the null interaction would require additional constraints on how alternatives contribute presuppositions to exhaustification.

Implications for interface architectures Taken together, the preceding discussion makes it difficult to reconcile the Type A systems with our observed absence of interaction between exclusivity and filtering, while the Type B systems (EXH¹ combined with any semantic theory for presupposition projection) remain the most direct fit to the data. At the same time, EXH¹ has been argued to face independent empirical challenges (e.g., with stress-sensitive inferences and phenomena discussed under “presupposed ignorance”; Spector & Sudo, 2017), suggesting a broader tension for fully compositional accounts that attempt to compute both scalar strengthening and presuppositional definedness conditions within a single semantic derivation.

A natural response is to consider shifting one of these computations to pragmatic reasoning, but our results do not by themselves support any particular pragmatic resolution. A pragmatic derivation of exclusivity (e.g., Gricean reasoning with the conjunction alternative) is still alternative-sensitive and must specify how presuppositions of alternatives are handled; if the conjunction alternative is treated in a way that imports its presuppositions as global constraints, pragmatic strengthening can replicate an EXH²-style feedback route and again predict reduced filtering under increased exclusivity, contrary to what we observe. Conversely, if presupposition projection is derived pragmatically (e.g., Schlenker’s theory based on the notion of transparency,⁶ or more general RSA-style theory of projective content), it is not guaranteed that projection will be independent of the availability of exclusive readings, especially if exclusivity is computed semantically. The more conservative conclusion, then, is architectural: whatever the division of labor between semantics and pragmatics, the interaction between strengthening and projection must be constrained so that (i) strengthened meaning does not systematically feed presupposition projection (mitigating the feed-forward route), and (ii) alternative-based reasoning does not systematically reimpose global presuppositions via presupposition-bearing alternatives (mitigating the feedback route), at least for the disjunction configurations tested here.

6. Conclusion

This paper asks how the exclusive interpretation of disjunction, often modeled as a scalar implicature, affects presuppositions projecting across disjunction. We combine (i) a theoretical survey showing that many fully compositional accounts of both exclusive interpretation and presupposition projection predict that stronger exclusivity should increase projection, with (ii) a Mandarin experiment testing this prediction that manipulates the availability of exclusive readings via environmental monotonicity. The experiment found no evidence that increasing the exclusivity of disjunction changes presupposition projection across it. If proven to be robust, this result is in tension with a broad class of semantic implementations (many semantic theories of projection combined with bivalent exhaustification or trivalent exhaustification based on strong negation) and motivates new work on interface constraints for how scalar implicatures interact with presupposition projection.

⁶ In fact, Mayr and Romoli (2016b) has shown that Schlenker’s theory (2009) based on the notion of transparency would make the Type A prediction, when the exclusive reading of disjunction is generated in semantics.

References

- Chierchia, G. and S. McConnell-Ginet (2000). *Meaning and Grammar: An Introduction to Semantics*. MIT Press.
- Chierchia, G., D. Fox, and B. Spector (2012). The grammatical view of scalar implicatures and the relationship between semantics and pragmatics. In P. Portner, C. Maienborn, and K. von Stechow (Eds.), *Semantics: An International Handbook of Natural Language Meaning* 3, 2297–2331. Berlin: Mouton de Gruyter.
- Fox, D. (2007). Free choice and the theory of scalar implicatures. In U. Sauerland and P. Stateva (Eds.), *Presupposition and Implicature in Compositional Semantics*, 71–120. Basingstoke: Palgrave.
- George, B. R. (2008). A new predictive theory of presupposition projection. *Semantics and Linguistic Theory* 18, 358–75.
- Heim, I. (1983). On the projection problem for presuppositions. In M. Barlow, D. Flickinger and N. Wiegand (Eds.), *Proceedings of WCCFL* 2, 114–25.
- Kalomoiros, A. and F. Schwarz (2024). Presupposition projection from “and” vs “or”: Experimental data and theoretical implications. *Journal of Semantics* 41(3-4), 331-72.
- Loder, M., J. Kuhn, and B. Spector (2024). Experimentally assessing the symmetry of presupposition filtering across disjunction. *Proceedings of NELS* 54, 51-60.
- Mandelkern, M., J. Zehr, J. Romoli, and F. Schwarz (2020). We’ve discovered that projection across conjunction is asymmetric (and it is!). *Linguistics and Philosophy* 43, 473–514.
- Marty, P. and J. Romoli (2021). Presuppositions, implicatures, and contextual equivalence. *Natural Language Semantics* 29, 229–280.
- Mayr, C. and J. Romoli (2016a). Satisfied or exhaustified: An ambiguity account of the Proviso Problem. *Semantics and Linguistic Theory* 26, 892-912.
- Mayr, C. and J. Romoli (2016b). A puzzle for theories of redundancy: Exhaustification, incrementality, and the notion of local context. *Semantics and Pragmatics* 9(7), 1–48.
- Rothschild, D. (2011). Explaining presupposition projection with dynamic semantics. *Semantics and Pragmatics* 4(3), 1–43.
- Russell, B. (1905). On denoting. *Mind* 14(56), 479-493.
- Schlenker, P. (2009). Local contexts. *Semantics and Pragmatics* 2(3), 1–78.
- Spector, B. and Y. Sudo (2017). Presupposed ignorance and exhaustification: How scalar implicatures and presuppositions interact. *Linguistics and Philosophy* 40, 473–517.
- Stalnaker, R. (1973). Presuppositions. *Journal of Philosophical Logic* 2(4), 447-457.
- Strawson, P. F. (1950). On referring. *Mind*, 59(235), 320-344.
- Tonhauser, J., D. Beaver, and J. Degen (2018). How projective is projective content? Gradience in projectivity and at-issueness. *Journal of Semantics* 35(3), 495-542.
- Wilcox, E. G., R. P. Levy, and K. Davidson (2021). Which presuppositions are subject to contextual felicity constraints? *Semantics and Linguistic Theory* 31, 345-364.
- Wilcox, E. G. (2022). *Informative presupposition & accommodation*. Ph.D. thesis, Harvard University.